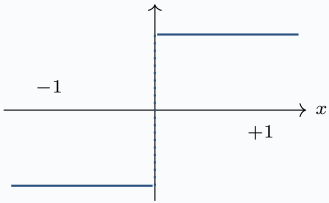
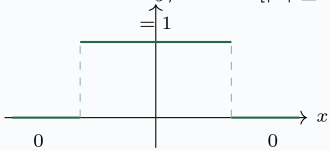
Aux A. Straight-Through Estimator ($\widetilde{\mathrm{sign}}$)

Forward: $y = \mathrm{sign}(x)$



Backward: $\partial y/\partial x = \mathbb{1}[|x| \le 1]$



Forward is hard binarisation; backward is identity on the active strip $[-1, 1]$ and zero outside. The continuous shadow $\mathbf{V}$ is updated by backpropagation; the recurrence always sees $\widetilde{\mathbf{V}} \in \{-1, +1\}^{V\times d}$.

Aux B. Deployment equivalent — XOR + popcount

Encode $-1 \mapsto 0$, $+1 \mapsto 1$. Then for $a, b \in \{-1, +1\}^d$:

$$\langle a, b\rangle \;=\; d - 2 \cdot \mathrm{popcount}(\bar{a} \oplus \bar{b}).$$

$\bar{a}$	1	0	1	1	0	1	0	1
$\bar{b}$	1	1	1	0	0	1	1	0
$\bar{a} \oplus \bar{b}$	0	1	0	1	0	0	1	1

popcount $= 4$
$d - 2 \cdot 4 = 0$
(orthogonal)

The 6502 inner loop: XOR two $d/8$-byte rows, popcount the result, subtract $2\cdot$popcount from d. No multiplications, no floating point, no NaN reachable. For $d = 256$: 32 bytes XORed and one popcount lookup per byte.